

WATERWALKER · DESIGN DECISION RECORD

WaterWalker recommendations accepted

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DOCUMENT CONTROL

Document ID	WWK-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/waterwalker		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Items 7, 8 and 14 recorded as decided by Amish on 2026-10-02
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); record the newly decided items, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted (items 9 to 13 of WWK-DDR-001); items 7, 8 and 14 decided by Amish on 2026-10-02 (WWK-DEC-001)

■ Context

WWK-DDR-001 decided the six TRL 2 review items and left seven items open. Five of them (items 9 to 13) came out of the TRL 3 calculations (WWK-CAL-001 v0.1) and each carried a recommendation; the other two (items 7 and 8) had none. The review note (`docs/REVIEW.md` , session 2026-09-25) also listed the budget for the later assist prototype with no recommendation.

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." This record applies that instruction to WaterWalker. Items with a recommendation are now decided; items without one stay open. TRL 4 remains on hold by Amish's instruction, and nothing here goes past TRL 3.

■ Decision

Each row in Table 1 is "Decided by Amish, 2026-09-25: go with recommendation".

Table 1. Newly decided items and what changed in the repo.

#	Item	Decision	What changed in the repo
9	Rear wheel mounting	Decided by Amish, 2026-09-25: go with recommendation. Each rear wheel sits in a fixed rigid bicycle fork with 100 mm dropouts and a 100 mm front-type drum hub; track 770 mm	Already in the model and drawing; status changed from proposed to decided in WWK-PRC-001 v0.4 and WWK-CAL-001 v0.2
10	Caster arm and front riser position	Decided by Amish, 2026-09-25: go with recommendation. Risers at $x = 1,150$ mm, wheelbase 1,530 mm, cradle 780 mm long	Already in the model and drawing; status changed to decided
11	Main frame section	Decided by Amish, 2026-09-25: go with recommendation. 40 x 30 mm rectangular tube for the side rails, front risers and caster arms	Section kept. Its wall is set by item 12 (1.2 mm), which is the later and more specific recommendation on the same member
12	Mass (R9)	Decided by Amish, 2026-09-25: go with recommendation. Apply the thinner cradle and the 1.2 mm main-tube wall, keep the puncture protection, and revisit R9 with users in co-design	<code>cad/src/model.py</code> : main tube 40 x 30 x 1.5 mm to 40 x 30 x 1.2 mm; cradle floor 9 to 6 mm, walls 6 to 4 mm, pad 4 to 2 mm. Empty mass 40.8 to 37.4 kg; loaded 125.2 to 121.8 kg. BOM items 1 (\$60 to \$55) and 5 (\$24 to \$20). Revisiting R9 with users waits for co-design (item 7)
13	Parking brake	Decided by Amish, 2026-09-25: go with recommendation. Add a positive lock pin; keep the lever latch for short stops	New BOM item 15 (\$5, 0.10 kg): a 10 mm pin pushed from the walking space through the left skirt guard and the left rear spokes, on a tab welded to the inner fork blade, projecting nothing beyond the axle nuts. Modelled in <code>cad/src/model.py</code> ; WWK-REQ-001 R10 restated

Effect on the numbers (WWK-CAL-001 v0.2)

Table 2. Before and after.

Quantity	v0.1	v0.2
Empty mass (with assist kit)	40.8 kg (47.3 kg)	37.4 kg (43.9 kg)
Push force, firm level path	24.6 to 61.4 N	23.9 to 59.7 N
Push force, 10 % climb	146.7 to 183.4 N	142.6 to 178.3 N
Push force, loose sand	242 to 368 N	236 to 358 N
Caster arm root stress at 2.5 g	112 MPa (factor 2.09)	133 MPa (factor 1.77)
Weld stress range at the rear hanger	63 MPa	75 MPa, above the 71 MPa detail category
Clay pot spare length	8 mm	12 mm
First-prototype parts	\$430	\$426

Requirement status changes: R2 and R4 move from not met to at risk; R11 moves from at risk to met (\$24 margin); R9 stays not met (2.4 kg over). Summary: 3 met, 7 at risk, 2 not met (R3 and R9), 1 not verifiable at TRL 3.

Budget

`budget_usd` stays at 450. No recommendation changed the budget figure or what it covers; R11 already covers the unassisted first prototype with mounting points (WWK-DDR-001 item 2).

Items on hold (TRL 4)

Revisiting R9 with users, and checking weld fatigue and lock pin fit, need co-design sessions, a build or tests. They are decided in direction but on hold, because TRL 4 is on hold by Amish's instruction.

■ Items left open on 2026-09-25 (decided 2026-10-02)

No recommendation was given on 2026-09-25; Amish decided all three on 2026-10-02 (WWK-DEC-001).

#	Item	Status
7	First co-design partner and region	Decided 2026-10-02: Helpful Engineering network as the route; Kitui County, Kenya, as the first candidate region
8	Dead-man brake	Decided 2026-10-02: a hold-to-release brake on the first prototype, kept with the lever, latch and lock pin; users asked in co-design
14	Budget for the later assist prototype (kit \$260, or about \$450 with two motors, plus a shared SwapCell pack)	Decided 2026-10-02: value-engineering target of about USD 450 for the two-motor kit, pack excluded

New finding for Amish, not decided: the thinner main-tube wall raises the weld stress range at the cradle hangers to 75 MPa, above the 71 MPa of a typical fillet-welded detail at 2 million cycles (WWK-CAL-001 v0.2 section 7). Options are gussets at the hanger and caster arm joints, a local 1.5 mm wall at those joints, or accepting the risk until a TRL 4 fatigue test. This is recorded in `docs/REVIEW.md` for Amish and is not applied.

■ Consequences

- `cad/src/model.py`, the STEP and STL exports, drawing WWK-DWG-001 (Rev P1 to P2), the concept media, `bom/bom.csv` and WWK-CAL-001 (v0.1 to v0.2) now reflect items 12 and 13.
- WWK-PRC-001 and WWK-REQ-001 move to v0.4 and WWK-DDR-001 to v0.2 with the decided status.
- No other repo is affected by these items.
- TRL stays at 3 (`trl: 3`, `trl_target: 3`). TRL 4 is on hold by Amish's instruction.